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Abstract:

Business Collaboration Platforms like Microsoft Teams and Slack enable teamwork by supporting text chatting and third-party resource integration. A user can access online file storage, make video calls, and manage a code repository, all from within the platform, thus making them a hub for sensitive communication and resources. The key enabler for these productivity features is a third-party application model. We contribute an experimental security analysis of this model and the third-party apps. Performing this analysis is challenging because commercial platforms and their apps are closed-source systems. Our analysis methodology is to systematically investigate different types of interactions possible between apps and users. We discover that the access control model in these systems violates two fundamental security principles: least privilege and complete mediation. These violations enable a malicious app to exploit the confidentiality and integrity of user messages and third-party resources connected to the platform. We construct

proof-of-concept attacks that can: (1) eavesdrop on user messages without having permission to read those messages; (2) launch fake video calls; (3) automatically merge code into repositories without user approval or involvement. Finally, we provide an analysis of countermeasures that systems like Slack and Microsoft Teams can adopt today.

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